

Machines C - Machines - Monta Vista Science Olympiad (Div. C) - 11-14-2025
Student Answer Sheet (/monta-vista/SM/AnswerSheet?tid=000GF1)

Instructions (shown before students start the test)

Welcome to Machines! You will have 50 minutes to complete this test. Please show work for all short answer questions and assume $g=9.8\text{m/s}^2$.

DEVICE TESTING: Masses for testing will be sent in the coaches' confidential document. Linked here is the submission form. (<https://forms.gle/CXjBeR2oMQfAUAfHA>) Device portion can be submitted anytime through the MVSOI testing window. If you choose to do device testing, please submit a google folder with a full continuous video of device testing. Video should include:

- Team name & number
- Up to 5 minutes allotted to device setup with a clear timer (start the timer when setup time is started)
- A clear view of device timer during Device Test Time.
- Clear displayed device testing w/ test mass adjustment and testing in line with the Rules Manual. Clear obtainment of estimated mass ratios & other data.

Good luck and please email me (elainaypan@gmail.com) for any questions!

Introduction (shown after students start the test)

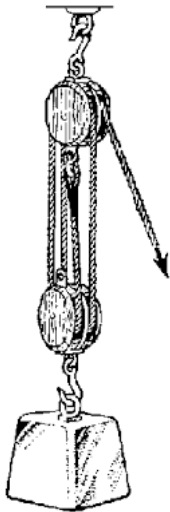
Welcome to Machines! You will have 50 minutes to complete this test. Please show work for all short answer questions and assume $g=9.8\text{m/s}^2$.

DEVICE TESTING: Masses for testing will be sent in the coaches' confidential document. Linked here is the submission form. (<http://Linked here is the submission form.https://forms.gle/CXjBeR2oMQfAUAfHA>) Device portion can be submitted anytime through the MVSOI testing window. If you choose to do device testing, please submit a google folder with a full continuous video of device testing. Video should include:

- Team name & number
- Up to 5 minutes allotted to device setup with a clear timer (start the timer when setup time is started)
- A clear view of device timer during Device Test Time.
- Clear displayed device testing w/ test mass adjustment and testing in line with the Rules Manual. Clear obtainment of estimated mass ratios & other data.

Good luck!

1. (2.00 pts) What type of pulley is this?



- ☐ A) Gun Tackle
- ☐ B) Gyn Tackle
- ☐ C) Twofold Purchase
- ☐ D) Threefold Purchase

2. (2.00 pts) What is the mechanical advantage of a Single Luff Tackle pulley?

- ☐ A) 1
- ☐ B) 2
- ☐ C) 3
- ☐ D) 4

3. (2.00 pts) A stapler is a ____ class lever.

- ☐ A) First
- ☐ B) Second
- ☐ C) Third

4. (2.00 pts) Scissors are a ____ class lever.

- ☐ A) First
- ☐ B) Second
- ☐ C) Third

5. (2.00 pts) What type of gear is this?



- ☐ A) Miter Gear
- ☐ B) Worm Gear
- ☐ C) Helical Gear
- ☐ D) Spur Gear

6. (2.00 pts) A wheelbarrow is a ____ class lever.

- ☐ A) First
- ☐ B) Second
- ☐ C) Third

7. (2.00 pts) A nutcracker is a ____ class lever.

- ☐ A) First
- ☐ B) Second
- ☐ C) Third

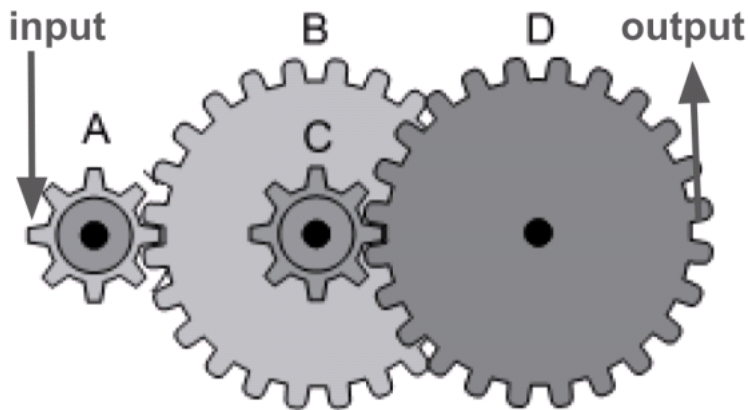
8. (3.00 pts) Find the mechanical advantage of an equilateral wedge.

- ☐ A) $\sqrt{3}$
- ☐ B) $1/\sqrt{3}$
- ☐ C) $\sqrt{3}/2$
- ☐ D) $2/\sqrt{3}$

9. (2.00 pts) A compound system has 5 simple machines, each with a mechanical advantage of 2. Find the total mechanical advantage of the system.

- ☐ A) 2
- ☐ B) 10
- ☐ C) 25
- ☐ D) 32

10. (3.00 pts) Gear A and C have 8 teeth, while gear B and D have 24 teeth. Find the mechanical advantage.



- ☐ A) 3
- ☐ B) $1/3$
- ☐ C) 9
- ☐ D) $1/9$
- ☐ E) 24
- ☐ F) $1/24$

11. (3.00 pts) A double start screw has a lead of 6mm. What is its pitch length?

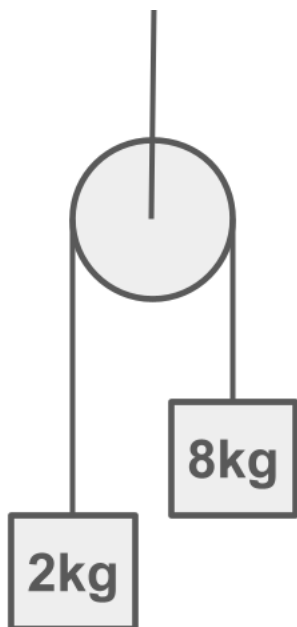
- ☐ A) 2mm
- ☐ B) 3mm
- ☐ C) 6mm
- ☐ D) 12mm

12. (3.00 pts) What is the mechanical advantage of the bottle opener?



- ☐ A) $\frac{1}{3}$
- ☐ B) 3
- ☐ C) 4
- ☐ D) 6

You set up a pulley mechanism shown below. The pulley is massless and frictionless.



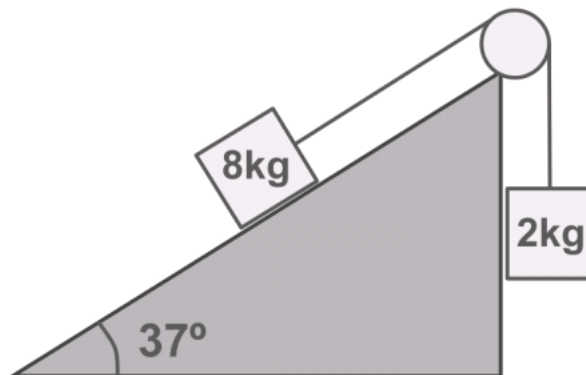
13. (3.00 pts) Find the acceleration of the blocks.

- ☐ A) $0.25g$
☐ B) $0.4g$
☐ C) $0.6g$
☐ D) $0.8g$

14. (4.00 pts) Find the tension in the string.

- ☐ A) 19.6N
☐ B) 31.4N
☐ C) 49.0N
☐ D) 78.4N

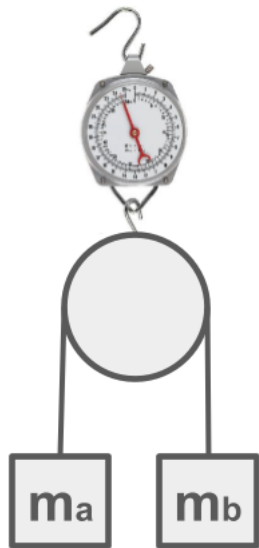
15. (4.00 pts) You decide this setup is too boring and moves her pulley onto a ramp. The ramp is frictionless.



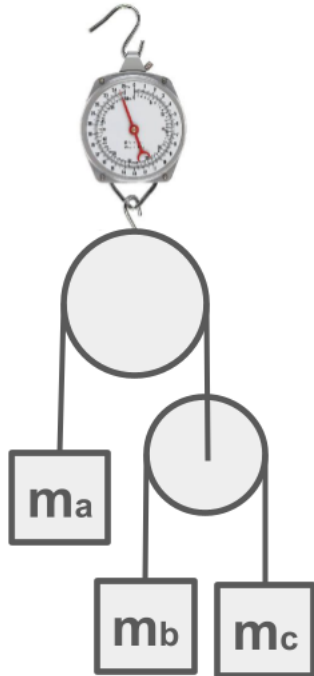
The 8kg block is 2.5m from the bottom of the ramp. How long after release does it take to get to the bottom of the ramp?

16. (5.00 pts)

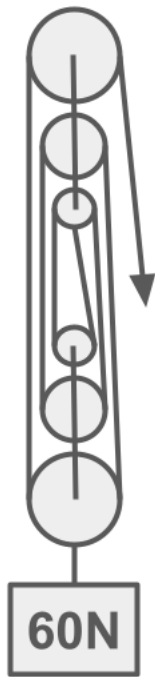
The system shown below consists of a scale, a pulley, and two hanging masses ma and mb . The pulley and string are massless and frictionless. Derive an expression for the weight reading on the scale in terms of ma and mb .



17. (5.00 pts) Mass m_b is replaced with a second pulley with hanging masses m_b and m_c . Derive an expression for the weight reading on the scale in terms of m_a , m_b , and m_c .

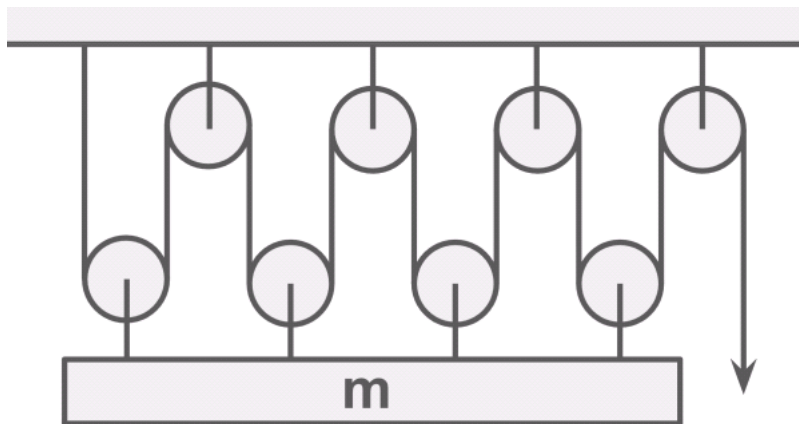


18. (3.00 pts) What is the minimum force needed to lift the weight?



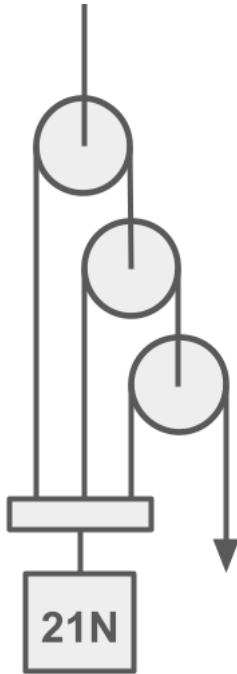
- ☐ A) 10N
- ☐ B) 12N
- ☐ C) 30N
- ☐ D) 60N

19. (4.00 pts) Find the mechanical advantage of the system.



- ☐ A) 2
- ☐ B) 4
- ☐ C) 8
- ☐ D) 16

20. (4.00 pts) The 21N weight is pulled up with a constant velocity. What is the minimum force needed to do so?



- ☐ A) 3N
- ☐ B) 7N
- ☐ C) 15N
- ☐ D) 21N

21. (4.00 pts) A 200N load is lifted with a gyn tackle pulley with 50N of effort. What is the efficiency of the system?

- ☐ A) 25%
- ☐ B) 40%
- ☐ C) 80%
- ☐ D) 100%

22. (3.00 pts) The coefficient of friction for a block of sand is $\mu = 0.4$. At what maximum angle can the sides of the sand pile remain stable?

- ☐ A) 18°
- ☐ B) 22°
- ☐ C) 24°
- ☐ D) 28°

23. (3.00 pts)

A compound machine is composed of a lever and wedge. For the lever, the distance between fulcrum and load is 0.9m, while distance between fulcrum and effort is 0.3m. The wedge has length 0.12m and thickness 0.04m. What is the mechanical advantage?

- ☐ A) 1

- ☐ B) 3
- ☐ C) 4.5
- ☐ D) 9

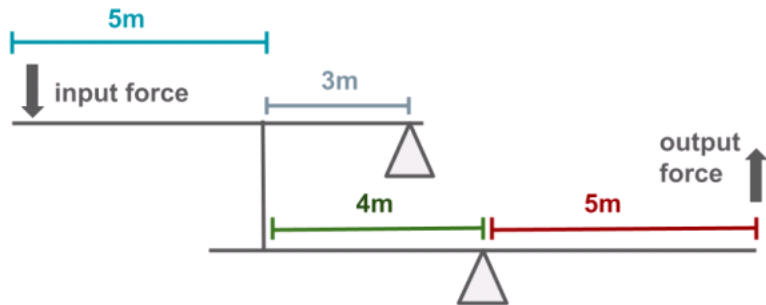
24. (3.00 pts)

A lever is used to lift a rock. The fulcrum is placed 0.5 m from the rock, and a 200 N force is applied 2m from the fulcrum on the opposite side. If the rock weighs 500 N, what is the efficiency of the lever?

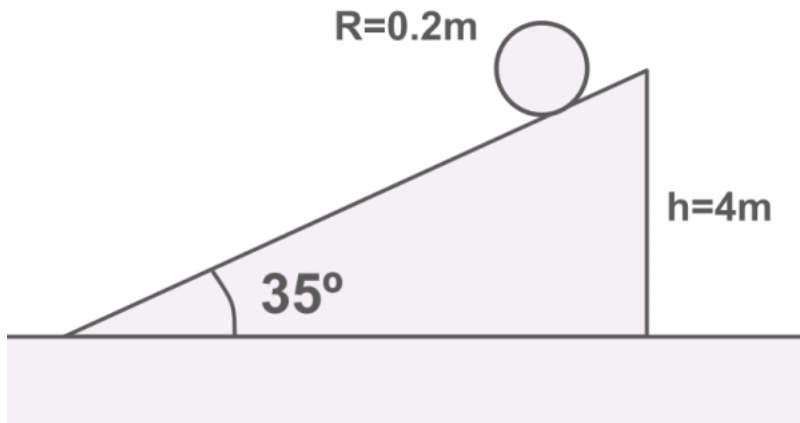
- ☐ A) 25%
- ☐ B) 50%
- ☐ C) 62.5%
- ☐ D) 80%

25. (2.00 pts) A class 3 lever will always have an IMA that is ____ than 1.

- ☐ A) greater
- ☐ B) less

26. (4.00 pts) Find the mechanical advantage of the compound lever shown below.

A sphere ($I = \frac{2}{5}mR^2$) with mass 1kg and radius 0.2m is placed on top of a ramp with height 4m and mass 10kg. The sphere rolls without slipping on the ramp. The ramp is placed on a surface with friction, keeping it in place.



27. (4.00 pts) What is the sphere's angular velocity when it reaches the bottom?

28. (5.00 pts) What is the normal force between the ramp and the ground when the sphere is rolling?

29. (4.00 pts)

The sphere is placed on a different, frictionless ramp with the same dimensions. The ramp is also on a frictionless surface. Find the velocity of the sphere once it exits the ramp.

A 1200kg car drives on a banked curve, angled at a 39° angle with radius $R = 50\text{m}$. The curve has a coefficient friction $\mu = 0.2$.

30. (4.00 pts) Find the car's minimum velocity when it is not slipping.

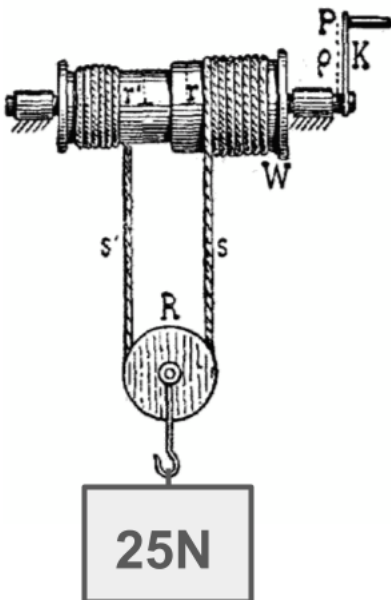
31. (4.00 pts) Find the car's maximum velocity when it is not slipping.

32. (4.00 pts) A screw has a radius of 2 cm and a handle with radius 20 cm. The lead angle of the screw is 5° . What is the ideal mechanical advantage (IMA) of the screw?

- ☐ A) 20
- ☐ B) 114
- ☐ C) 229
- ☐ D) 458

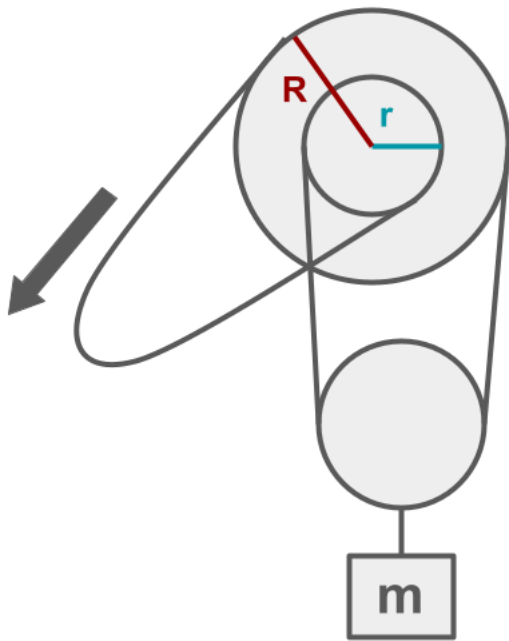
33. (4.00 pts)

The differential windlass below has left drum $R = 0.10\text{m}$ and right drum $r = 0.08\text{m}$ with opposite winding. The handle's length is 0.25m . Find the minimum force applied on the handle to raise the load.



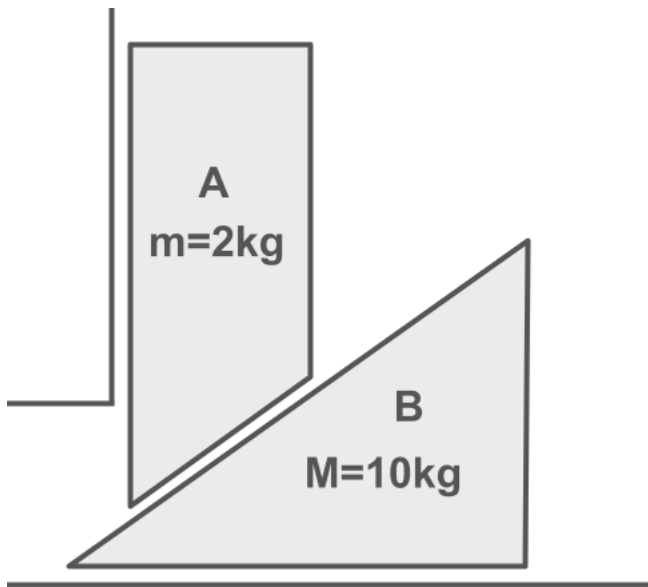
- ☐ A) 1N
- ☐ B) 2N
- ☐ C) 4N
- ☐ D) 10N

34. (4.00 pts) Derive an expression for the force needed to pull the mass m up at a constant velocity, for a differential pulley in terms of R and r .

**35. (7.00 pts)**

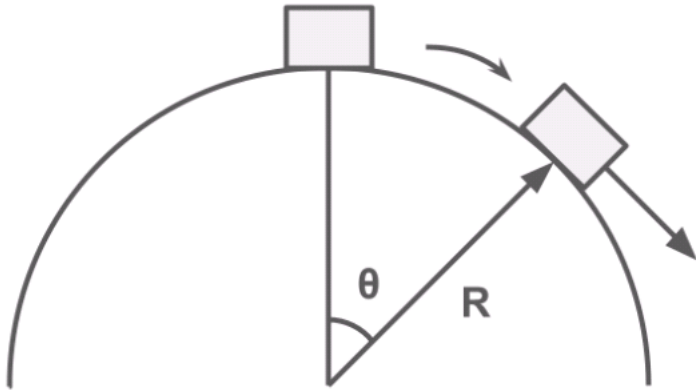
A small block A of mass $m=2.0\text{kg}$ rests against a smooth inclined face of a movable wedge B of mass $M=6.0\text{kg}$. The wedge has an incline angle of 30° and rests on the ground. The upper block rests against a vertical rail and can only move up and down.

Initially, both blocks are at rest, and the blocks are released at $t=0$. Find the displacement of block A after $t=1$ second.

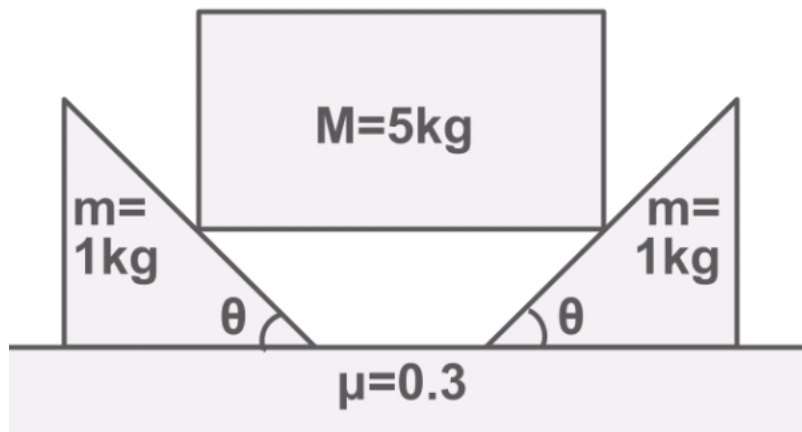


36. (6.00 pts)

A block starts at rest at the top of a large frictionless hemisphere of radius $R=10\text{m}$ and slides, eventually losing contact with the surface. Find the speed of the block at the instant it leaves the surface.

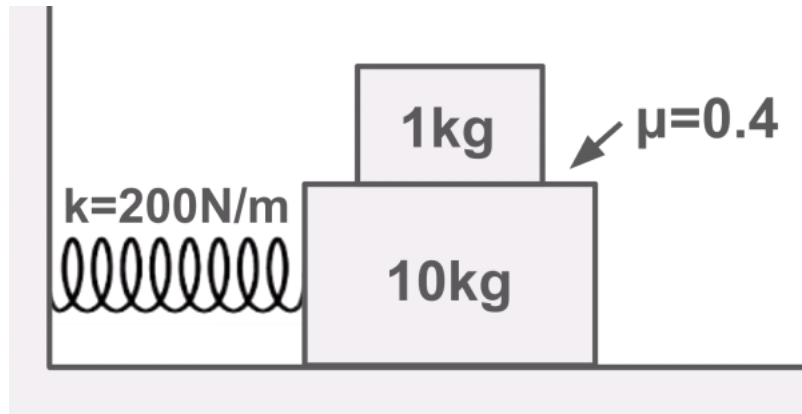
**37. (7.00 pts)**

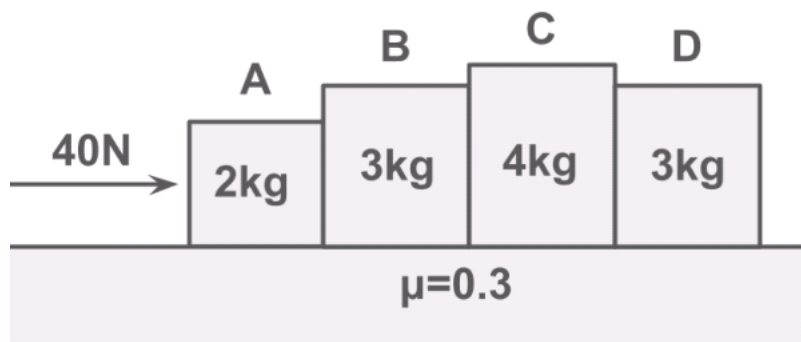
A rectangular block of mass $M=5\text{ kg}$ rests symmetrically between two identical smooth triangular ramps, each of mass $m=1\text{ kg}$. The ramps are placed on a horizontal surface that has a coefficient of static friction $\mu = 0.30$. There is no friction between the rectangular block and the surfaces of the ramps, and the system is in static equilibrium. Find the maximum ramp angle θ .



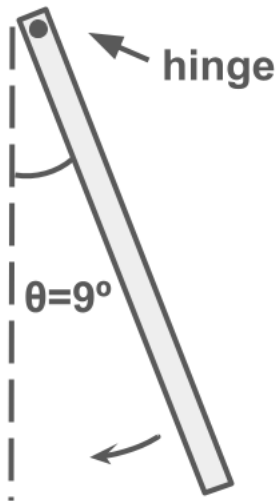
38. (7.00 pts)

Two blocks and a spring are arranged on a horizontal, frictionless surface as shown. The coefficient of static friction between the two blocks is 0.40. If the smaller block is not to slip, what is the maximum possible amplitude of oscillation?



39. (4.00 pts) Find the normal force between blocks B and C.

A hanging rod hinged at its top is released from a 9° angle from the vertical. The hinge is frictionless and the rod has mass M .



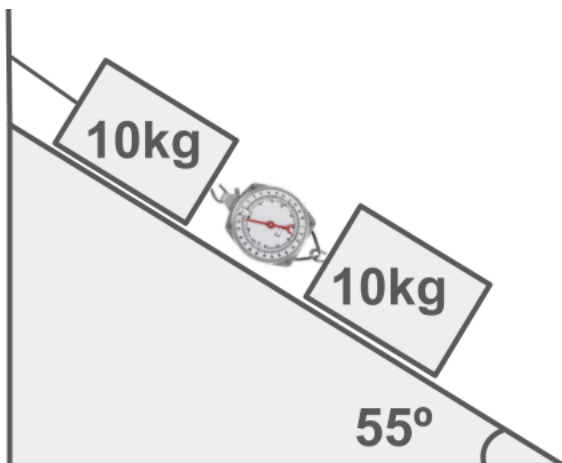
40. (5.00 pts) What is the angular acceleration of the rod when released?

- ☐ A) $g \sin \theta / L$
☐ B) $g \sin \theta / 2L$
☐ C) $2g \sin \theta / L$
☐ D) $3g \sin \theta / 2L$

41. (6.00 pts)

Find the magnitude of the hinge force acting on the rod at the instant it is released, expressed in terms of the rod's weight Mg (For example, if the hinge force equals half the rod's weight, write the answer as 0.500Mg.)

42. (5.00 pts) The masses shown below are placed on a frictionless 55° ramp. What is the force on the scale reading?



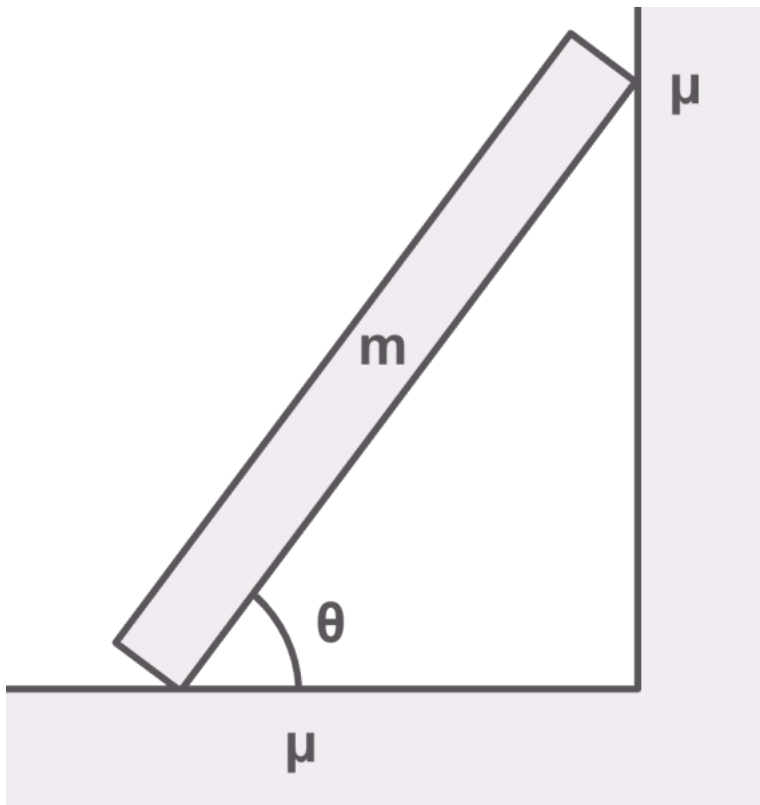
- ☐ A) 0N
- ☐ B) 80N
- ☐ C) 100N
- ☐ D) 160N

43. (3.00 pts) You roll a sphere, a cylinder, and a hoop, all with identical radii and masses, off a ramp. Which one will reach the bottom fastest?

- ☐ A) Sphere
- ☐ B) Cylinder
- ☐ C) Hoop

44. (6.00 pts)

A uniform ladder rests against a rough wall and a rough floor, both with coefficient of static friction μ . Find the minimum angle $\theta_{\min}$ with the ground required for the ladder to remain in equilibrium.



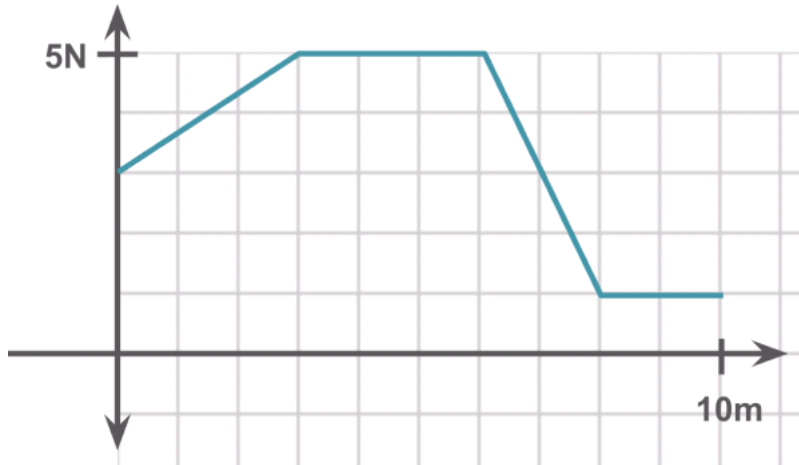
- ☐ A) $\arctan((1-\mu^2)/\mu)$
- ☐ B) $\arctan((1+\mu^2)/\mu)$
- ☐ C) $\arctan((1-\mu^2)/2\mu)$
- ☐ D) $\arctan((1+\mu^2)/2\mu)$

45. (2.00 pts) A screw jack will be self-locking if:

- ☐ A) $\sin\lambda < \mu$

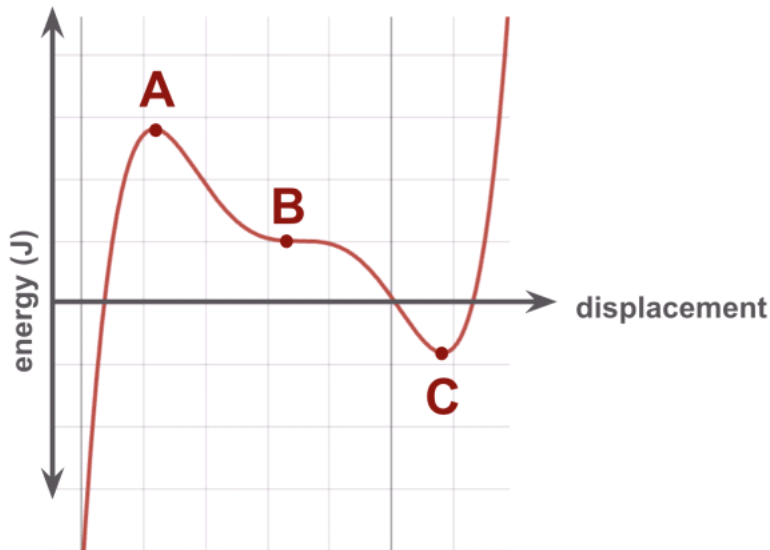
- ☐ B) $\cos\lambda < \mu$
- ☐ C) $\tan\lambda < \mu$

46. (5.00 pts) A 1 kg object is initially at rest and is acted upon by a variable force as shown in the graph below (force vs. displacement). What is its final velocity?



47. (4.00 pts)

The graph shows how the potential energy (U) of a particle varies with displacement (x). The points A, B, and C represent equilibrium positions. Which of the following correctly identifies the type of equilibrium at each point?



- ☐ A) A: stable, B: unstable, C: neutral
- ☐ B) A: stable, B: neutral, C: unstable
- ☐ C) A: unstable, B: neutral, C: stable
- ☐ D) A: unstable, B: stable, C: unstable

48. (4.00 pts) A compound pulley supports 5 rope segments. If each rope has 5% friction loss, find effective MA.

- ☐ A) 3.75
- ☐ B) 3.86
- ☐ C) 4.75
- ☐ D) 4.88

49. (5.00 pts) A student uses a ramp to lift a 300N box. The student applies an effort of 100 N, and the ramp is 75% efficient. What is the angle of the ramp?

- ☐ A) $\arcsin(1/3)$
- ☐ B) $\arcsin(1/4)$
- ☐ C) $\arcsin(3/4)$
- ☐ D) $\arcsin(4/9)$

50. (4.00 pts) Two students lift objects vertically through the same height. Student A lifts a 10-kg mass in 5 seconds, while Student B lifts a 20-kg mass in 10 seconds. What is the ratio of Student A's power output to Student B's (P_A/P_B)?

- ☐ A) 1/2
- ☐ B) 1
- ☐ C) 2
- ☐ D) 4

Thank you for taking the test and good luck for your other events! Please email me (elainaypan@gmail.com) for any questions.

Submission Link: <https://forms.gle/CXjBeR2oMQfAUAfHA> (<https://forms.gle/CXjBeR2oMQfAUAfHA>)